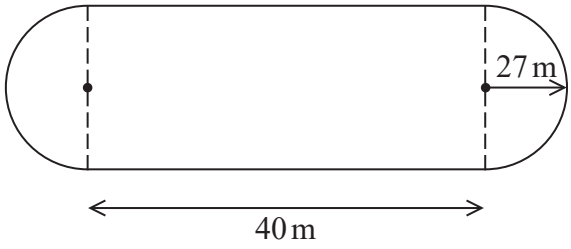
26 The diagram shows a cycle track.



The track has two straight sides each of length 40 m.
Each end of the track is a semicircle of radius 27 m.

The diameter of each wheel of Ian's bike is 590 mm.
Ian is going to ride his bike around the track once.

Calculate how many complete revolutions each wheel of his bike will make.

Two semicircular ends make one full circle.

Curved distance = $2\pi \times 27 = 54\pi$ m.

1 mark
Both ends

Track = $80 + 54\pi = 249.6460033...$ m.

1 mark
Add two straights

Wheel diameter = 590 mm = 0.590 m.

Wheel circumference = $0.590\pi = 1.853539666...$ m.

1 mark
Wheel circumference

Revolutions = $(80 + 54\pi) \div (0.590\pi) = 134.686...$

1 mark
Same units; divide

Each wheel makes134 complete revolutions.

1 mark
134; round down

(Total for Question 26 is 5 marks)